

STT Studio

Self-Hosted, Real-Time Speech-to-Text on NVIDIA DGX Spark

A marketing & content briefing pack for the GenAI Protos marketing team

Product Overview — Elevator Pitch

STT Studio is a production-minded, **fully on-premise speech-to-text platform** that runs two best-in-class open-source ASR models — **Qwen3-ASR-1.7B** and **OpenAI Whisper-Large-v3** — side-by-side on a single NVIDIA DGX Spark. It ships as a one-command Docker stack with a friendly browser UI, an OpenAI-compatible REST API, and a low-latency WebSocket for live microphone transcription.

It is purpose-built for organizations that need **high-quality multilingual transcription** — **especially Modern Standard Arabic** — **without sending a single byte of audio to the public cloud.**

One-liner: *Live, multilingual, Arabic-grade speech-to-text that runs entirely on your own GPU — set up in one command.*

Key Features

Live Microphone Transcription

- Speak into the browser, see the transcript appear in **under 1.5 seconds** end-to-end.
- Built on `getUserMedia` → `AudioWorklet` → **WebSocket** streaming of raw PCM16 audio.
- Server-side **Silero VAD** (Voice Activity Detection) intelligently chunks speech on natural silence boundaries — no clipped words, no runaway segments.

File Upload Transcription

- Drag-and-drop any audio/video file through the UI, or POST it to a REST endpoint.
- OpenAI-compatible `/v1/audio/transcriptions` — drop-in replacement for cloud APIs in existing code.

True Multilingual Support

- **30 languages** via Qwen3-ASR (SOTA for Arabic, Chinese, and many low-resource languages).
- **99 languages** via Whisper-Large-v3 (broadest coverage, mature ecosystem).
- Right-to-left aware UI — Arabic script renders correctly out of the box.

Dual-Model Side-by-Side Comparison

- Switch between Qwen3 and Whisper with a single click in the UI.
- Benchmark **Word Error Rate (WER)** across both on your own audio samples.
- Built-in failover — if one model struggles on a dialect or accent, the other is one flag away.

Zero Data Leaves the Box

- Runs **100% locally** on your hardware — no outbound calls after first-time model download.
- Ideal for healthcare, legal, defense, government, finance, and any regulated environment.
- CORS-scoped, localhost-bound by default.

OpenAI-Compatible API

- Point the OpenAI Python or Node SDK at `localhost:8000` — it just works.
- Migrating from the OpenAI Whisper API? **Change one URL, keep your code.**

Built-in Observability & Benchmarking

- `/healthz`, `/readyz`, `/metrics` endpoints for liveness, readiness and Prometheus-style metrics.
- Structured JSON logs (structlog) — pipe directly into ELK, Loki or Datadog.
- `scripts/bench.sh` runs an end-to-end **WER benchmark** on your own audio samples.

One-Command Deploy

- `bash scripts/setup.sh && bash scripts/start.sh` — that's it.
 - Three containers (Qwen vLLM + Whisper vLLM + FastAPI backend) orchestrated by Docker Compose.
 - Hot-reload dev mode for engineers, lean prod mode for operators.
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Technical Highlights

Layer	Technology	Why it matters
Primary ASR model	Qwen3-ASR-1.7B (Alibaba, Apache 2.0)	State-of-the-art open-source ASR, streaming-capable, 30 languages including Arabic MSA.
Secondary ASR model	Whisper-Large-v3 (OpenAI, MIT)	Industry benchmark, 99 languages, mature tooling.
Inference serving	<code>vLLM</code> × 2 (NGC <code>26.03-py3</code>)	Production-grade LLM serving with paged attention, CUDA graphs, OpenAI-compatible HTTP API.
Backend	FastAPI + Starlette + Pydantic v2 + httpx + structlog	Async, type-safe, low-overhead routing between browser and vLLM.
Voice Activity Detection	Silero VAD v5 (MIT)	Server-side chunking on 500 ms silence after ≥250 ms speech, with 15 s max segment — tuned for Arabic utterance lengths.
Streaming protocol	WebSocket + raw PCM16 @ 16 kHz mono	No WebRTC, no proprietary protocol — simple, debuggable, firewall-friendly.
Audio capture	Browser <code>AudioWorklet</code> (Float32 → Int16)	Runs on the audio rendering thread — glitch-free, no main-thread jank.
Hardware	NVIDIA DGX Spark (GB10 Blackwell, sm_121, aarch64)	128 GB unified memory, ~8 GB VRAM consumed by both models combined — massive headroom.
Container runtime	NVIDIA vLLM container <code>26.03-py3</code>	First NGC tag with native <code>Qwen3ASRForConditionalGeneration</code> support on Blackwell.
Orchestration	Docker Compose v2 + Makefile + shell scripts	Reproducible, no Kubernetes cognitive tax for single-node deployments.
Dev-experience	<code>uv</code> , <code>ruff</code> , <code>mypy</code> , <code>pytest</code> , smoke tests, WER bench	Enforced lint + types + tests — production-grade Python hygiene.

Clever engineering calls worth highlighting in a technical post

1. **Chunk-on-silence instead of continuous streaming.** Rather than fight the known Qwen3-ASR realtime-endpoint quality bug, we chunk on natural silence boundaries via VAD and send each utterance as a fresh transcription call. Result: *sub-1.5 s perceived latency with full-utterance quality* — no proprietary ML streaming code required.

2. **Two models, one GPU.** Both models fit comfortably together; switching is a config flag, not a redeploy.
 3. **Blackwell / sm_121 pitfalls solved.** The Dockerfiles encode non-obvious fixes: Flash-Attention backend for Whisper (Triton lacks encoder-decoder cross-attention on sm_121), avoiding PyPI torchaudio (ABI-incompatible with NVIDIA's FP4 torch), vLLM patch for processor imports. **This alone saves a new team ~2 weeks of debugging.**
 4. **OpenAI wire-compatible from day one.** Any client that speaks the OpenAI audio API — Python SDK, Node SDK, curl, Postman — works unmodified against either model.
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Benefits

For CTOs & Architects

- **Cloud-cost independence** — no per-minute transcription fees, predictable capex.
- **Data sovereignty** — fully compliant with on-prem and air-gapped requirements (HIPAA, GDPR Article 44, UAE PDPL, KSA PDPL).
- **Vendor neutrality** — open models, open licenses (Apache 2.0 / MIT), no lock-in.

For Product & Engineering Teams

- **Drop-in OpenAI replacement** — migrate existing Whisper API code in minutes.
- **Reproducible stack** — same stack on a developer laptop (with a GPU) as in production.
- **Benchmark-ready** — built-in WER tooling lets teams pick the right model *for their own audio*, not generic leaderboards.

For Operations

- **One-box deployment** — a single DGX Spark runs the whole thing.
- **Healthchecks + metrics + structured logs** — observability built in, not bolted on.
- **Low maintenance surface** — three containers, no Kubernetes, no managed-service dependencies.

For End Users

- **Real-time captions in the browser** — click, speak, read.
 - **Accurate Arabic transcription** — MSA and major dialects handled with care.
 - **No uploads to third parties** — private by design.
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Suggested Taglines & Headlines

- *"Arabic-grade speech-to-text, entirely on your own GPU."*

- "One box. Two models. Ninety-nine languages. Zero data leakage."
- "Real-time transcription with the quality of a cloud API and the privacy of a local network."
- "From silence to subtitle in under 1.5 seconds — on a single DGX Spark."

Architecture Diagram — Image-Generation Prompt

Paste this into your image-gen model of choice (Midjourney, DALL·E 3, Imagen, Flux, Stable Diffusion with a technical LoRA):

Prompt:

A clean, modern enterprise software architecture diagram, isometric 3D perspective, flat vector style with soft gradients, dark navy and teal color palette with orange accent highlights, white background, sharp technical illustration.

Top layer: a web browser window labeled 'Browser UI' showing a microphone icon, a waveform, and a live transcript in both English and Arabic (right-to-left). An arrow labeled 'WebSocket + REST (port 3000)' flows downward.

Middle layer: a single rounded rectangle labeled 'FastAPI Backend' containing four small icons — a microphone (WebSocket /ws/transcribe), an upload arrow (POST /transcribe), a heart-pulse (healthz/readyz/metrics), and a soundwave with scissors (Silero VAD chunker). Two arrows flow downward from it, labeled 'http://qwen:8000' and 'http://whisper:8001'.

Lower-middle layer: two parallel glowing server-rack rectangles side by side. The left one labeled 'vLLM Server #1 — Qwen3-ASR-1.7B, 30 languages, Arabic-primary'. The right one labeled 'vLLM Server #2 — Whisper-Large-v3, 99 languages'. Both display 'OpenAI-compatible /v1/audio/transcriptions'.

Bottom layer: a large stylized NVIDIA DGX Spark box with green GPU lighting, labeled 'NVIDIA DGX Spark · GB10 Blackwell · 128 GB Unified Memory · sm_121 · aarch64'. Both vLLM servers sit on top of it, sharing the GPU.

Around the whole stack: a subtle dashed boundary labeled 'On-Premise — No Data Leaves The Box' with a padlock icon in the corner.

Typography: clean sans-serif (Inter-like), generous whitespace, technical but friendly. Style: Stripe-docs / Linear-blog / Vercel aesthetic. No cartoon characters, no people, no clutter.

Tip for the designer: if the image-gen model struggles with the layered layout, generate it in two passes — first the DGX Spark base layer with both vLLM servers on top, then composite the Browser → FastAPI top portion in Figma.

Content Angles — Post Ideas for the Marketing Team

1. **"Why we run Qwen3-ASR and Whisper side-by-side"** — dual-model architecture, benchmarking story.
 2. **"Getting Arabic speech-to-text right: what the cloud APIs miss"** — Arabic / MSA quality angle.
 3. **"Deploying vLLM on DGX Spark Blackwell: the 5 gotchas that cost us a week"** — developer blog, SEO goldmine.
 4. **"Replacing the OpenAI Whisper API in your product without changing a line of client code"** — migration playbook.
 5. **"Real-time, sub-1.5-second browser transcription — no WebRTC, no magic, just VAD and WebSockets"** — engineering deep-dive.
 6. **"Data sovereignty in speech AI: why regulated industries are bringing ASR back on-prem"** — compliance / thought-leadership angle.
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